

Executive Summary

Proof of Concept - Predicting Jira Story Points

This document summarizes the results of a proof-of-concept project aimed at developing a machine learning model to predict Jira issue Story Points based on the free-text content of their 'Detailed Description'.

Utilizing a small dataset of 200 Jira issues, the project successfully trained a Random Forest Regressor model after vectorizing the text descriptions using TF-IDF. The model was evaluated on an unseen test set (40 issues) and demonstrated exceptionally high predictive performance:

- Mean Absolute Error (MAE): 0.1167 Story Points
- Root Mean Squared Error (RMSE): 0.1799 Story Points
- R-squared (R²): 0.9943

These metrics indicate that the model accurately predicts Story Points with very low average error and explains nearly all the variance in Story Points within this dataset.

Furthermore, SHAP (SHapley Additive exPlanations) analysis was employed to provide crucial model explainability. This allowed for the identification of the most influential words and terms in the 'Detailed Description' that drive predictions, both overall and for specific issues. SHAP plots visually demonstrated how particular words contribute to increasing or decreasing the predicted Story Points.

To transition this analysis into a practical tool, the trained model, the text vectorizer, and the SHAP explainer were saved. A function was implemented to load these components and enable predictions on new, user-provided detailed descriptions. This function not only returns the predicted Story Point value but also generates a SHAP explanation plot, offering insight into the why behind the specific prediction for the new input.

In conclusion, this proof-of-concept successfully demonstrated the feasibility of predicting Jira issue Story Points from text descriptions with very high accuracy on the available dataset. The integration of SHAP explainability provides valuable transparency, showing which terms influence the estimates. This capability offers a strong potential for assisting or partially automating the initial story point estimation process, providing data-driven predictions with interpretable reasoning. Future work would benefit from testing performance on larger and more diverse datasets.